Akash Prajapati

Python Backend Engineer | Applied AI & LLM Systems

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PROFESSIONAL SUMMARY

Python Backend Engineer with 3.4+ years of experience developing scalable backend systems, REST APIs, and microservices using Python, Django, DRF, and FastAPI. Experienced in database optimization, asynchronous processing, and enterprise integrations, with hands-on exposure to Applied AI, RAG pipelines, LLM integrations, and semantic search across automotive, e-commerce, and AI-driven applications.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, Django, Django REST Framework, FastAPI, REST APIs, Microservices

Applied AI & GenAI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Tool Calling, OpenAI APIs, Semantic Retrieval, Image Embeddings, Sentence Transformers, ChromaDB, Whisper, ElevenLabs, Guardrails, LLM Harnessing, LLM Evaluation, LangSmith

Databases & Messaging: PostgreSQL, Redis, Django ORM, Celery, Apache Kafka

Authentication & Integrations: Keycloak, JWT, OAuth 2.0, OTP Verification, Payment Gateway Integration, Third-Party API Integration, RBAC

Concepts: OOP, Database Design, Query Optimization, Agile Methodology, System Design

Cloud & Tools: AWS (S3, RDS, EC2, ELB), Docker, Kubernetes (Fundamentals), Grafana, Git, GitHub, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Vishleshan Software Solutions Pvt. Ltd. | Software Developer 25th November 2025 – 24th September 2026

- Developed enterprise e-commerce workflows using Python, Django, DRF and AWS RDS PostgreSQL, contributing to 100+ REST APIs supporting 1,000+ daily orders; integrated DSR/MSR, Shiprocket and SAP Grow workflows.
- Developed FastAPI-based notification services integrated with Apache Kafka, supporting approximately 1,000–10,000 daily notifications; implemented Redis caching for order history and optimized Django ORM queries.
- Built asynchronous reporting, bulk export and scheduled processing workflows using Celery and Redis, integrating AWS S3 for file storage and secure presigned URL-based downloads.
- Developed AI-powered product search using separate image and text embedding pipelines, enabling discovery through images, part numbers, product names, categories and voice queries, with top-5 similarity-based retrieval.
- Containerized the search application using Docker, provisioned an AWS EC2 instance and deployed the service in a cloud environment.
- Implemented a RAG-based Hindi Voice AI assistant using Sentence Transformers, ChromaDB, recursive document chunking, semantic retrieval, cosine similarity, query rephrasing and fallback handling; orchestrated workflows using LangGraph conditional routing.
- Integrated Whisper, GPT-4 and ElevenLabs for speech-to-text, LLM-based response generation and text-to-speech; implemented LLM tool calling to execute backend API operations and trigger email-sending workflows.
- Improved LLM application reliability through PII redaction, structured output validation and prompt injection detection; used LangSmith for execution tracing, debugging and monitoring latency and token usage.
- Evaluated AI response quality across approximately 5,000 test cases using expected answers and LLM-based evaluation, achieving an average score of approximately 87%.
- Implemented authentication and API security using Keycloak, JWT, OAuth 2.0 and OTP verification; contributed to production issue resolution and backend support.

Sankey Solutions | Solution Analyst 1st June 2023 – 14th November 2025

- Managed and contributed to approximately 18 interconnected microservices built using Python, Django and PostgreSQL on AWS RDS, supporting vehicle management, fleet operations, subscriptions and payment workflows.
- Built vehicle tracking functionality to process historical GPS coordinates, calculate distances using the Haversine formula, identify trips and stops, and prepare route data for Google Maps visualization.
- Improved API performance by 20–30% by resolving N+1 query issues and optimizing database access using Django ORM Subquery and query-level optimizations.
- Implemented role-, site- and subscription-based access controls, managing Fleet Owner and Fleet Manager permissions, site-level vehicle visibility and plan-based access.
- Developed mileage analytics and business logic using driving parameters such as speed, idling, harsh acceleration, braking and gear shifting/RPM to generate mileage-improvement recommendations.
- Implemented historical telemetry synchronization through third-party APIs using 30-minute scheduled jobs, checkpoint-based incremental fetching, backlog recovery and PostgreSQL upserts keyed by VIN and timestamp.
- Integrated the CCAvenue payment gateway with request encryption, response decryption, payment callbacks, status updates and subscription synchronization; implemented subscription purchase, upgrade, renewal and expiry workflows.
- Designed database structures and developed 50+ REST APIs using Python, Django and PostgreSQL for enterprise quality management, digitizing vehicle assembly-line inspections, defect reporting and corrective action workflows.

- Implemented Quality Maker and Quality Checker workflows and developed an Apache Kafka consumer to ingest vehicle information from manufacturing sites into PostgreSQL; integrated Swagger/OpenAPI documentation.
- Independently developed an AI-assisted Git repository analysis solution integrating GitHub APIs and OpenAI to summarize code changes and identify potential security vulnerabilities, including exposed API keys, tokens, passwords and AWS credentials; generated structured reports with issue descriptions and severity levels.

EDUCATION

Bachelor of Engineering — Computer Engineering
Smt. Indira Gandhi Engineering College, University of Mumbai | Aug 2019 – Jun 2023 | CGPA: 8.88/10

ACHIEVEMENTS

- **Shining Star:** Recognized for outstanding technical contributions.
- **Bar Raiser:** Recognized for problem-solving, technical guidance, and supporting team members.

CERTIFICATIONS

- Machine Learning with Google Cloud
- Android Development

PUBLICATION

- Crop Protection Against Birds Using Deep Learning and IoT, IRJET, Vol. 10, Issue 4, April 2023.